In-Situ Biochar Amendment for Bioretention

[Bracketed text in red should be adjusted or removed per region and project goals. Guidance for biochar-specific decisions is provided in the Biochar in Stormwater BMPs Guidebook for Stormwater Practitioners (LINK).]

PART 1 - GENERAL

Description

This specification outlines the process for amending biochar and sand into existing soils, covering furnishing, soil preparation, and amendment applications. The process integrates biochar and sand to enhance stormwater infiltration, permeability, and bulk density while accounting for excess soil removal due to bulk density changes.

Purpose

Proper incorporation of biochar materials into existing (in-situ) soils can increase infiltration and water holding capacity and increase moisture and soil nutrient retention to improve urban plant growth and success. To ensure performance as guided by the research, all the bioretention media must have at least 60% sand by volume and less than 20% clay.

Related Specifications

Biochar Material and Quality Control
Biochar Material Handling
Biochar Material Stockpiling
Bioretention Media [from regional guidance and/or standard, or specify below]
Compost [from regional guidance and/or standard]

References

International Biochar Initiative. 2015. Standardized Product Definition and Product Testing Guidelines for Biochar That Is Used in Soil Version 2.1 [IBI-STD-2.1]. Table 1. (LINK)

Brady, N. C., & Weil, R. R. (2008). The nature and properties of soils (14th ed.). Pearson Prentice Hall. (Table 1)

Soil Survey Staff. 2022. Keys to Soil Taxonomy, 13th ed. USDA-Natural Resources Conservation Service. (LINK).

ASTM International. (2017). ASTM D2487-17: Standard practice for classification of soils for engineering purposes (Unified Soil Classification System). ASTM International. (LINK).

U.S. Department of Agriculture (USDA) Natural Resources Conservation Service (NRCS). (2017). National Engineering Handbook Part 630: Hydrology (Chapter 7 – Hydrologic Soil Groups). U.S. Government Printing Office. (LINK).

Rawls, W. J., Brakensiek, D. L., & Saxton, K. E. (1982). Estimating soil water retention and hydraulic properties from soil texture. Soil Science Society of America Journal, 46(5), 1138-1144.(LINK).
Rawls, W. J., Giménez, D., & Grossman, R. (1998). Use of soil texture, bulk density, and slope of the water retention curve to predict saturated hydraulic conductivity. Transactions of the ASAE, 41(4), 983-988. (LINK).

Wong, C., Wilkins, A., Fraley-McNeal, L., & Hegberg, C. 2025. Biochar as an Enhancement to Bioretention Practices in the Chesapeake Bay Watershed. Center for Watershed Protection (CWP) and Infinite Solutions. USWG White Paper.

Submittals

1. Submit per requirement under [General Provisions, Section XXX, ...]
2. Testing results from in-situ soil test, including particle size distribution, organic matter content, bulk density, saturated hydraulic conductivity, and infiltration rate.

Delivery, Storage, and Handling

Refer to the following specifications

1. Biochar Material Stockpiling
2. Biochar Material Handling

PART 2 - MATERIALS

Biochar

Refer to the following specifications

1. Biochar Material and Quality Control

Biochar shall account for [15% to 20%] by volume of the finished in-situ bioretention mix.

Sand

1. Sand shall follow ASTM C33 gradation for medium to coarse construction sand.
2. Sand shall be washed to ensure fines and dust are not present.

PART 3 – EXECUTION

Application Methods

1. Application Rate of biochar for soil amendment shall be 15% to 20% by volume with remaining soil composition adjustments made with sand to align with HSG A per Figure XX. By adjusting the remaining soil composition with sand, the soil structure, hydraulic conductivity, and infiltration performance will align with the desired HSG classification.

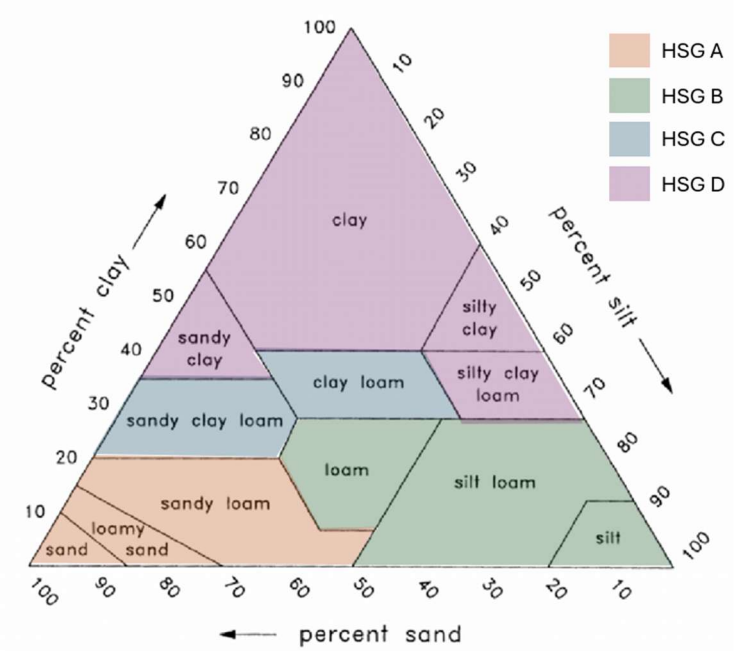


Figure 1. Soil texture and hydraulic soil group based on particles size distribution

3. Avoid over-mixing biochar to prevent increasing fines and breaking down the biochar particle size, which could negatively impact soil porosity and infiltration performance.
4. Depth of Integration shall be [6-12 inches for turf cover, 6-24 inches minimum for perennial and hardwood plantings].
5. The spreading and mixing of biochar and final stabilization shall be completed in a one-day operation or ≥5,000 sf with proper Erosion & Sediment Control (E&SC) permits and authorization to prevent the loss of biochar material from wind, rainfall, or runoff.
6. Complete the installation of each modified soil amendment area from top of slope to bottom and at no time shall more than one treatment area be open at the same time as another.

Sequence of Construction

1. Do not complete this work when soil moisture is greater than 40%.
2. Remove turf and overburden prior to beginning soil preparation work.
3. To avoid damaging in-situ soil structure by digging and turning unless being removed to blend in biochar and/or sand.
4. **Shallow Application (<12 inches) [Use as applicable]**
   a. Subsoil (yeoman or similar) the treatment area to a depth of 24 inches.
   b. Once subsoiling is completed, rotary power harrow (or other similar tool approved by Scaling Up Biochar project) shall be used to till the freshly subsoiled soil to a depth of 11 inches.
   c. Using the rotary power harrow [or other approved tool], till in the biochar at a speed to allow for a through mixing into the native soil.

d. After the first pass of the rotary harrow, spread and rake out 1/3 of total project biochar uniformly. Use a drop spreader or similar to uniformly distribute biochar over project area.
e. Once all the required biochar has been thoroughly tilled into the in-situ soils, rake out any tracks and level out any high areas to minimize potential for concentrated runoff or ponding water.
f. Repeat this process (step 4d) twice before all project biochar is used.
g. Seed, straw, and tackify the disturbed area prior to disturbing any additional area. [Coir peat matting can be used as an alternative.] [Seed variety to be determined by property owner or planting schedule.]
h. In any area that could have concentrated water flow should use core logs to break up concentrated flows and velocities avoiding erosion issues.
i. Once modified soil amendment area has been stabilized and approved by the designated project consultant, the next soil amendment area can be started.

5. **Deep Application (>12 inches) [Use as applicable]**
   a. Remove soil to the desired depth based on modification requirements.
   b. Blend biochar into the soil mixture at [15%-20%] by volume to improve infiltration and porosity.
   c. Mix sand at the required percentage by volume with the in-situ soil to adjust soil texture.
   d. Replace the amended soil in 12-inch lifts, ensuring even distribution throughout the soil profile.
   e. If feasible, subsoil using a single or double shank at the bottom elevation before replacing the mixed soil to improve natural infiltration and break up any restrictive layers.
   f. Do not compact the soil during mixing or replacement to maintain permeability.
   g. Once all the required biochar has been thoroughly tilled into the in-situ soils, rake out any tracks and level out any high areas to minimize potential for concentrated runoff or ponding water.
   h. Seed, straw, and tackify the disturbed area prior to disturbing any additional area. [Coir peat matting can be used as an alternative.] [Seed variety to be determined by property owner or planting schedule.]
   i. In any area that could have concentrated water flow should use core logs to break up concentrated flows and velocities avoiding erosion issues.
   j. Once modified soil amendment area has been stabilized and approved by the designated project consultant, the next soil amendment area can be started.

6. **Biochar Moisture Management**
   a. Biochar may need to be moistened to reduce dust and prevent biochar floating or surface accumulation during mixing. Do not saturate biochar or soils.

b. Since biochar has a lower bulk density than native soils, the amendment process causes volume expansion, requiring excess soil removal. This ensures proper integration of amendments without increasing the final soil level.
c. Ensure proper soil moisture levels before final grading to prevent uneven settling.